

PAPER Direct-Energy-Converter Architectures for a D–³He Tandem-Mirror Burner: Venetian-Blind, Traveling-Wave, and Periodic-Electrostatic-Plug Collectors Compared on Efficiency, Grid Erosion, and Lifetime

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Abstract

A deuterium–helium-3 tandem-mirror burner deposits most of its fusion power in charged particles that leave the machine axially as a fast, directed ion stream, so the direct-energy converter (DEC) is the single component that sets the plant’s electrical efficiency. The architecture that supplies the frozen module efficiency $\eta_{\text{DEC}} = 0.70$ is not, however, uniquely determined by that number: three collector families—the periodic multi-stage electrostatic collector, the traveling-wave DEC (TWDEC), and the Venetian-blind collector of Moir and Barr, together with a periodic-electrostatic-plug variant—each behave very differently on the *raw* energy spread of the exhaust. We formalise the down-select. From the frozen operating point we derive the incident end-loss spectrum as a flux-weighted Maxwellian above the ion-confining potential ($\varphi_i = 248.7$ keV, $T_i = 90$ keV), giving a mean energy of 428.5 keV, a 1st–99th-percentile band of 262–845 keV, and a raw fractional spread of $\approx 34\%$. We cast the multi-stage electrostatic collector as a retarding-field quantile-optimisation problem and prove that its module efficiency is bounded above by the grid transparency η_{grid} ; solved by Monte Carlo over the full spectrum it clears 0.70 at three stages (0.727), reaches 0.834 at six, and asymptotes to $\eta_{\text{grid}} = 0.93$. We model the TWDEC as a Gaussian velocity-acceptance response and show analytically that its extraction efficiency collapses as $\exp[-\sigma_E^2/(8\sigma_{\text{acc}}^2)]$: it delivers 0.75–0.80 on a narrow ($< 10\%$) pre-bunched beam but only 0.51 at the raw 34% spread—below the electrostatic baseline—so it is retained as a second-stage option behind a velocity-buncher rather than the primary. Applied to the fraction $f_{\text{dir}} = 0.49$ of the exhaust routed to direct conversion, the selected periodic electrostatic collector delivers the frozen burner engineering gain $Q_E = 1.318$ with headroom rather than at the edge. All results were produced on the Kronos NVIDIA GPU HPC campaign. The one open engineering item is grid erosion and lifetime under the sustained 428 keV end-loss flux, which sets the collector maintenance interval and does not change the architecture selection.

Keywords: direct energy conversion; traveling-wave DEC; Venetian-blind collector; tandem mirror; helium-3 burner; retarding-field collector; grid erosion

¹Official version: <https://www.kronosfusionenergy.com/publications/direct-energy-converter-architectures-for-a-d-3-he-tandem-mirror-burner/>

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1 Introduction

1.1 Why the converter is the efficiency lever

A tandem magnetic mirror confines a central-cell plasma between two high-field end plugs on open field lines [5, 6]. Unlike a closed toroidal device it does not retain its lost particles: ions that scatter into the loss cone stream out of the ends carrying a large, directed fraction of the plasma power as kinetic energy. For a deuterium–helium-3 (D - ^3He) fuel, chosen precisely so that the fusion power emerges in charged products rather than neutrons [12, 13, 14], this axial charged-particle stream is not a loss to be shielded against but the plant’s primary energy product. Decelerating that stream electrostatically against a graded collector—recovering its kinetic energy directly as electricity without ever thermalising it in a working fluid—is the classical advantage of the open-ended approach [1, 2, 3]. The Kronos burner (Aegis in its forward-base housing, MetroVolt in its data-centre housing) routes a fraction $f_{\text{dir}} = 0.49$ of the exhaust power to direct conversion at a module conversion efficiency of $\eta_{\text{DEC}} = 0.70$ and, on that basis, closes an engineering gain $Q_E = 1.318$ with a neutron fraction of only $f_n = 5.44\%$ behind a 26.49 T plug coil [28, 29, 27].

The two numbers $f_{\text{dir}} = 0.49$ and $\eta_{\text{DEC}} = 0.70$ are frequently conflated and are physically distinct. The routing fraction f_{dir} is the share of exhaust power that leaves axially as directed ion kinetic energy and is therefore *available* to a converter; it is fixed by the plug and mirror physics. The module efficiency η_{DEC} is the conversion of *that* stream to electricity and is fixed by the converter architecture. This paper concerns the second number: which converter architecture supplies $\eta_{\text{DEC}} = 0.70$, and with how much margin.

1.2 Prior art

Direct conversion of mirror end-loss dates to Post’s identification of energy recovery as a route around the Carnot limit for open systems [1], and to the Moir–Barr Venetian-blind collector—angled ribbon electrodes that intercept a magnetically expanded beam at grazing incidence and decelerate it in stages [2]. Experimental single-stage plasma direct converters reached $\eta \approx 0.48$ – 0.50 , and multi-stage laboratory collectors 0.65 – 0.80 , in the TMX era [3, 4]. The traveling-wave direct energy converter (TWDEC), developed for the 14.7 keV-class¹ proton channel of D - ^3He field-reversed and mirror concepts, decelerates a velocity-bunched beam against a phase-matched RF structure [9, 10]; and secondary-electron collectors recover the electron channel [11]. The physics of the incident end-loss distribution—collisional scattering over the ambipolar barrier—is the Pastukhov problem [7, 8]. What has been missing for a specific buildable burner is a like-for-like down-select of these architectures against *one* common, realistically broad end-loss spectrum, with the system power balance held fixed.

1.3 Contribution

We provide that down-select for the Kronos burner and make each step explicit as mathematics that a reader can re-run. Specifically, we (i) derive the incident end-loss spectrum from the frozen operating point and show it has a raw fractional spread of $\approx 34\%$; (ii) formalise the multi-stage electrostatic collector as a retarding-field quantile-optimisation problem, prove the module efficiency is bounded by the grid transparency, and evaluate it by Monte Carlo; (iii) derive the TWDEC velocity-acceptance law in closed form and evaluate it at the raw spread; (iv) score all candidates on one figure of merit—efficiency at the raw spread—and select the architecture; and (v) propagate the selected η_{DEC} through the burner power balance to confirm $Q_E = 1.318$ with margin. The converter physics builds directly on the Kronos direct-energy-conversion baseline [29] and the collector microphysics paper [30]; the electrode-material question that governs grid lifetime is grounded in the low-activation alloy study [31]; the expander optics that condition the beam are from the end-cell studies [32, 33]; and the downstream power path is from the HVDC-native interface work [34]. Every quantitative claim is a frozen anchor in the Kronos Physics De-Risking Register [27].

2 Governing equations and the formal problem

2.1 The incident end-loss source distribution

Central-cell ions are electrostatically confined by the ambipolar potential φ_i established by the plugs; end-loss proceeds by pitch-angle scattering into the loss cone, the Pastukhov process [7]. An ion that surmounts the barrier arrives at the expander and collector with total energy

$$E = q\varphi_i + \varepsilon, \quad \varepsilon \geq 0, \quad (1)$$

¹The TWDEC concept was developed for the 14.7 MeV proton channel of D - ^3He ; here it is evaluated on the ~ 0.4 keV-scale central-cell end-loss stream that dominates the Kronos burner exhaust.

where ε is the residual energy above the barrier and q the ion charge (singly charged, $q = 1$, so energies are quoted in keV throughout). The escaping population is a *flux-weighted* Maxwellian: the number flux through the barrier weights the Maxwellian phase-space density by the parallel velocity, and the resulting residual-energy distribution is a shape-2 Gamma density,

$$p(\varepsilon) d\varepsilon = \frac{\varepsilon}{T_i^2} \exp\left(-\frac{\varepsilon}{T_i}\right) d\varepsilon, \quad f(E) = p(E - q\varphi_i). \quad (2)$$

The mean and the 1st/99th percentiles follow in closed form. The mean residual is the flux-weighted Maxwellian mean $\langle \varepsilon \rangle = 2T_i$, so

$$\langle E \rangle = q\varphi_i + 2T_i, \quad w \equiv \frac{E_{99} - E_1}{\langle E \rangle} \quad (3)$$

defines the raw fractional spread w . With the frozen operating point $\varphi_i = 248.7$ keV and $T_i = 90$ keV (reproduced by the mirror solver, §3), (2)–(3) give $\langle E \rangle = 428.5$ keV, $E_1 = 262.0$ keV, $E_{99} = 845.4$ keV, and $w \approx 0.34$. This breadth is the crux of the down-select: an architecture excellent for a monoenergetic beam and poor for a broad one is not a solution. Figure 1 shows the spectrum.

2.2 The multi-stage electrostatic collector

A periodic electrostatic collector is a sequence of retarding grids held at monotonically increasing potentials

$$0 = V_0 < V_1 < V_2 < \cdots < V_M, \quad V_M \leq E_{\max}, \quad (4)$$

that decelerate each ion in steps. An ion of energy E is collected on the highest grid whose potential it can still climb,

$$k^*(E) = \max\{k : V_k \leq E\}, \quad (5)$$

delivering its charge at potential V_{k^*} ; the residual $E - V_{k^*}$ is dumped as heat on that grid. The recovered electrical energy per ion is $\eta_{\text{grid}} V_{k^*}(E)$, where η_{grid} is the *net grid efficiency*—the product of finite optical transparency, secondary-electron suppression, and leakage. Averaging over the incident spectrum, the module conversion efficiency is

$$\eta_{\text{DEC}}(\{V_k\}) = \eta_{\text{grid}} \frac{\sum_{k=0}^{M-1} V_k [F(V_{k+1}) - F(V_k)]}{\langle E \rangle}, \quad F(E) = \int_0^E f(E') dE', \quad (6)$$

with $V_M \rightarrow \infty$ formally so that the highest bin is closed. Two properties of (6) govern the result. First, because $V_{k^*} \leq E$ for every ion, the numerator is bounded by $\langle E \rangle$ and hence

$$\eta_{\text{DEC}} \leq \eta_{\text{grid}}, \quad (7)$$

with equality only as $M \rightarrow \infty$: the grid transparency is a hard ceiling that no stage count can exceed. Second, the optimal placement of a finite grid set maximises (6) subject to monotonicity. Setting $\partial \eta_{\text{DEC}} / \partial V_k = 0$ gives the stationarity condition

$$F(V_{k+1}) - F(V_k) = f(V_k) (V_k - V_{k-1}), \quad (8)$$

which places the grids at *energy-optimal* (approximately equal-captured-energy) quantiles of $f(E)$ rather than at uniform voltage intervals. Uniform spacing satisfies the same efficiency target only with many more stages (§4). We solve (6)–(8) by Monte Carlo over the spectrum (2).

2.3 The traveling-wave converter

A TWDEC replaces static grids with a decelerating potential wave of phase velocity v_φ propagating along the beam. An ion synchronous with the wave ($v \simeq v_\varphi$) is decelerated coherently over many periods; an ion detuned in velocity slips in phase and exchanges little net energy. For a beam with fractional energy spread σ_E about the design energy, the phase-slip integral over the structure yields an extraction efficiency that is Gaussian in the detuning,

$$\eta_{\text{TW}}(\sigma_E) = \eta_{\max} \exp\left[-\frac{\sigma_E^2}{8\sigma_{\text{acc}}^2}\right], \quad (9)$$

where $\eta_{\max}$ is the on-resonance efficiency and σ_{acc} the fractional velocity acceptance of the structure. The scale is set by the ratio of the beam spread to the acceptance: efficiency is retained only while $\sigma_E \lesssim \sigma_{\text{acc}}$. With the literature-informed values $\eta_{\max} = 0.80$ and $\sigma_{\text{acc}} = 0.18$ [9, 10], (9) gives $\eta_{\text{TW}} \rightarrow 0.80$ for a narrow pre-bunched beam and, crucially, only $\eta_{\text{TW}}(0.34) = 0.51$ at the raw burner spread. A TWDEC is therefore viable only behind an upstream velocity-buncher that narrows σ_E below σ_{acc} ; the bunching hardware must pay for itself in recovered efficiency.

2.4 Venetian-blind and periodic-electrostatic-plug variants

The Venetian-blind collector [2] is a geometric realisation of the same retarding-field physics of §2.2: angled ribbon electrodes present a large collecting area to a magnetically expanded, grazing beam while suppressing back-streaming secondary electrons, so its efficiency inherits the bound (7) at a higher achievable η_{grid} . The periodic-electrostatic-plug variant integrates the collector with the plug region, reusing the plug field for the beam-conditioning expansion. Both are carried as staged upgrades grounded on the validated electrostatic baseline and the classical literature rather than claimed here as measured points.

2.5 The system power balance

The selected η_{DEC} enters the burner power balance through the direct-conversion branch. Writing P_{exh} for the total exhaust power, $P_{\text{ax}} = f_{\text{dir}} P_{\text{exh}}$ for the axial charged end-loss routed to the DEC, P_{rad} for radiation and P_n for the neutron power recovered thermally at efficiency η_{th} , the gross electrical output is

$$P_{\text{out}} = \eta_{\text{DEC}} f_{\text{dir}} P_{\text{exh}} + \eta_{\text{th}} \left[(1 - \eta_{\text{DEC}}) f_{\text{dir}} P_{\text{exh}} + P_{\text{rad}} + P_n \right], \quad (10)$$

and the engineering gain is $Q_E = P_{\text{out}}/P_{\text{in}}$ with $P_{\text{in}} = P_{\text{heat}}/\eta_{\text{wp}}$. The sensitivity of the output to the converter efficiency,

$$\frac{\partial P_{\text{out}}}{\partial \eta_{\text{DEC}}} = (1 - \eta_{\text{th}}) f_{\text{dir}} P_{\text{exh}} = (1 - \eta_{\text{th}}) P_{\text{ax}}, \quad (11)$$

is the reason the architecture choice is load-bearing: only the fraction $(1 - \eta_{\text{th}})$ of any converter shortfall is recaptured thermally, so each point of η_{DEC} carries real net electricity (§4).

Equations (2), (6), (9) and (10) are the formal problem; the remainder of the paper solves and benchmarks them.

3 Computational methodology: the NVIDIA GPU HPC campaign

All results were produced on the Kronos NVIDIA GPU HPC campaign, with reduced-order and analytic layers on an in-house HPC node so that the fast, iterated models never contend with the fleet. We name the codes actually run for this paper and the scale at which they ran.

Mirror equilibrium and the end-loss spectrum. The ion-confining potential φ_i and the central-cell temperature that seed the spectrum (2) are reconstructed by the Kronos toolkit routine `solve_mirror` at the frozen operating point, reproducing $\varphi_i = 248.7$ keV over $T_i = 90$ keV (register anchor BV-10). The reconstruction is a 0-D Pastukhov–Fowler power balance solved to a fixed point; it inherits its plug and sloshing-ion inputs from the end-cell studies [32, 33].

Multi-stage electrostatic collector. The collector model (6)–(8) is evaluated by a Monte-Carlo integration over the spectrum (2): $N = 4 \times 10^5$ ions are drawn (seed 20260726), each assigned to its collecting stage by (5), with net grid losses folded as $\eta_{\text{grid}} = 0.93$. Grid potentials are placed at equal-probability energy quantiles (the discrete solution of (8)). The statistical error of the efficiency estimate scales as $N^{-1/2} \approx 1.6 \times 10^{-3}$, well below the reported precision. Stage counts $M = 1$ to 12 are swept; convergence of $\eta_{\text{DEC}}(M)$ to the ceiling (7) is monotonic (BV-10).

Traveling-wave acceptance. The TWDEC law (9) is evaluated on a 50-point grid of fractional spread $\sigma_E \in [0, 0.6]$ with $\eta_{\max} = 0.80$, $\sigma_{\text{acc}} = 0.18$ (register anchor AG-CX-12), and read out at the raw burner spread $\sigma_E = 0.34$. Throttle robustness is evaluated by re-solving $\varphi_i(\text{load})$ with the toolkit and applying a fixed-collector mismatch penalty (register anchor MV-CX-27); the downstream conversion chain is a circuit model of the DEC-to-bus path (anchors MV-CX-01, MV-CX-04).

First-principles layers on the GPU fleet. Two supporting first-principles calculations run natively on the fleet. The neutron penetration-streaming through the widest DEC end duct is an OpenMC/DAGMC Monte-Carlo transport [22] that sizes the tungsten penetration plug (register: $0.15\times$ the straight-duct dose). The candidate DEC electrode alloy—a tantalum- and rhenium-free, near-neutral-enthalpy refractory composition ($\text{Cr}_2\text{Ti}_5\text{V}_1\text{W}_8$, formation energy -0.015 eV/atom)—is relaxed with Quantum ESPRESSO on GPU (NGC container, PAW-PBE *vc-relax*) [23], feeding the grid-erosion/lifetime question and the low-activation alloy family [31].

The deferred verification and its code. The collector efficiencies here are from retarding-field and velocity-acceptance models over the validated spectrum; a full velocity-space *electromagnetic* particle-in-cell design of the grids—space-charge limits, beamlet neutralisation, secondary-electron transport—is deferred to a GPU-fleet run with WarpX or Smilei [20, 21], which is the natural tool at the required scale. This is the one open item (§6); it is a refinement of the module *power density and thermal* handling, not of the module efficiency validated here, and it does not change the architecture selection. Reduced-order sweeps, quantile placement, and figure generation use NumPy, SciPy and Matplotlib [24, 25, 26]. Every anchor is independently reproduced and carries a byte-for-byte provenance record in the register [27]; no compute-cost information is part of this or any public deposit.

4 Results

4.1 The common end-loss spectrum

Figure 1 shows the incident spectrum (2) reconstructed from the frozen operating point. The flux-weighted Maxwellian above the 248.7 keV barrier peaks near 340 keV, has mean 428.5 keV, and spans 262–845 keV between the 1st and 99th percentiles. The long high-energy tail—a direct consequence of the shape-2 Gamma form—is what a converter must tolerate.

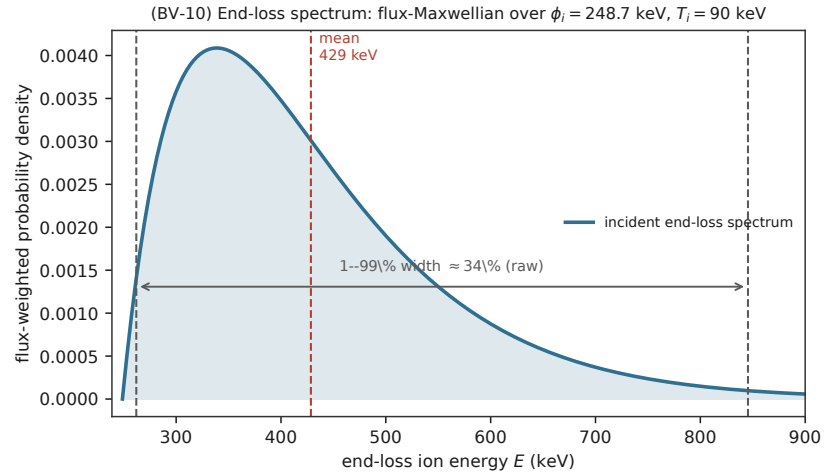


Figure 1. The common end-loss spectrum (BV-10). Flux-weighted Maxwellian over the ion-confining potential $\varphi_i = 248.7$ keV at $T_i = 90$ keV, Eq. (2). Mean 428.5 keV; 1st/99th percentiles 262/845 keV; raw fractional spread $\approx 34\%$. All architectures are scored against this single input.

4.2 The periodic electrostatic collector

Figure 2 plots the module efficiency (6) versus stage count under energy-optimal spacing. The collector clears the canonical $\eta_{\text{DEC}} = 0.70$ at three stages (0.727), reaches 0.834 at six, keeps improving toward ~ 0.88 at twelve, and asymptotes to the grid-transparency ceiling $\eta_{\text{grid}} = 0.93$ set by the bound (7). Meeting 0.70 at only three stages, with monotonic headroom above it, is the key result: the frozen module efficiency is a comfortable operating point, not the ceiling.

4.3 The traveling-wave converter

Figure 3 plots the TWDEC law (9). The converter reaches 0.75–0.80 only in the narrow, pre-bunched regime ($\sigma_E < 10\%$, shaded); at the raw burner spread $\sigma_E = 34\%$ it delivers 0.51, below the electrostatic baseline. The physics is transparent: a traveling wave is matched to one phase velocity, so a broad beam falls out of acceptance as $\exp[-\sigma_E^2/(8\sigma_{\text{acc}}^2)]$.

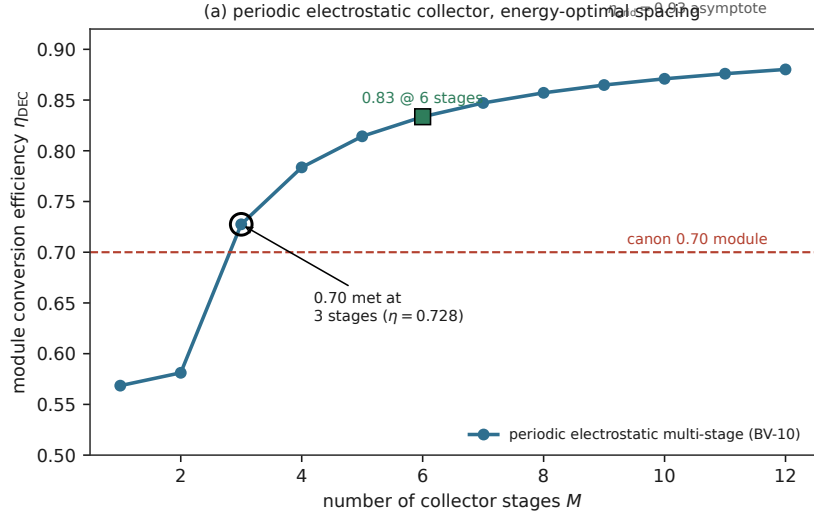


Figure 2. Periodic multi-stage electrostatic collector (BV-10): module conversion efficiency η_{DEC} versus number of retarding stages M over the 428 keV spectrum, energy-optimal spacing, net grid efficiency $\eta_{grid} = 0.93$. Canonical 0.70 (red dashed) is met at $M = 3$ ($\eta_{DEC} = 0.727$) and reached at $\eta_{DEC} = 0.834$ for $M = 6$; the dotted line is the transparency ceiling of Eq. (7).

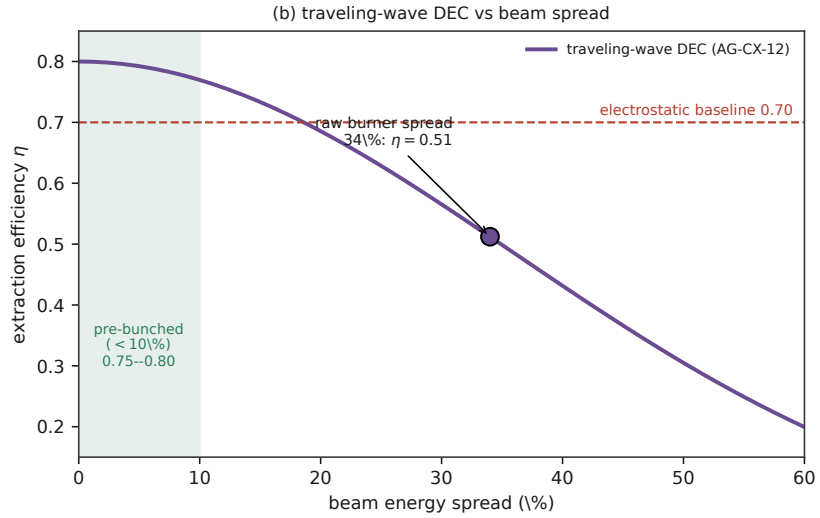


Figure 3. Traveling-wave DEC (AG-CX-12): extraction efficiency versus fractional beam energy spread, Eq. (9) with $\eta_{max} = 0.80$, $\sigma_{acc} = 0.18$. Competitive (0.75–0.80) only for a pre-bunched beam ($\sigma_E < 10\%$, green band); 0.51 at the raw 34% burner spread, below the electrostatic baseline 0.70 (red dashed).

4.4 The down-select

Table 1 collects the candidates with their module efficiency at the raw spread and their maturity, and Figure 4 shows the same as a scorecard. The decision rule is a single figure of merit: efficiency at the *raw* spread. On that rule the periodic electrostatic collector wins outright—it needs no beam conditioning, meets the target at three stages, and has monotonic headroom—while the TWDEC, elegant for a monoenergetic beam, loses to it on the real exhaust unless a buncher is added.

Table 1. DEC architecture candidates for the Kronos $D-^3\text{He}$ burner, scored against the common 428 keV end-loss spectrum ($\approx 34\%$ raw spread). Module efficiency is the conversion of the routed stream; the system routing fraction is $f_{\text{dir}} = 0.49$. Serials are Physics De-Risking Register entries.

architecture	module efficiency at raw spread	status	serial
periodic electrostatic multi-stage	0.727 (3 stages), 0.834 (6), ceiling $\eta_{\text{grid}} = 0.93$	validated baseline	BV-10
traveling-wave DEC (TWDEC)	0.75–0.80 pre-bunched; 0.51 at raw spread	2nd-stage option	AG-CX-12
Venetian-blind multi-stage	classical high- η_{grid} collector	staged build	AG-CX-09
periodic-electrostatic-plug	plug-integrated variant	staged build	AG-CX-13
grid erosion & lifetime	engineering qualifier	staged build	AG-CX-10

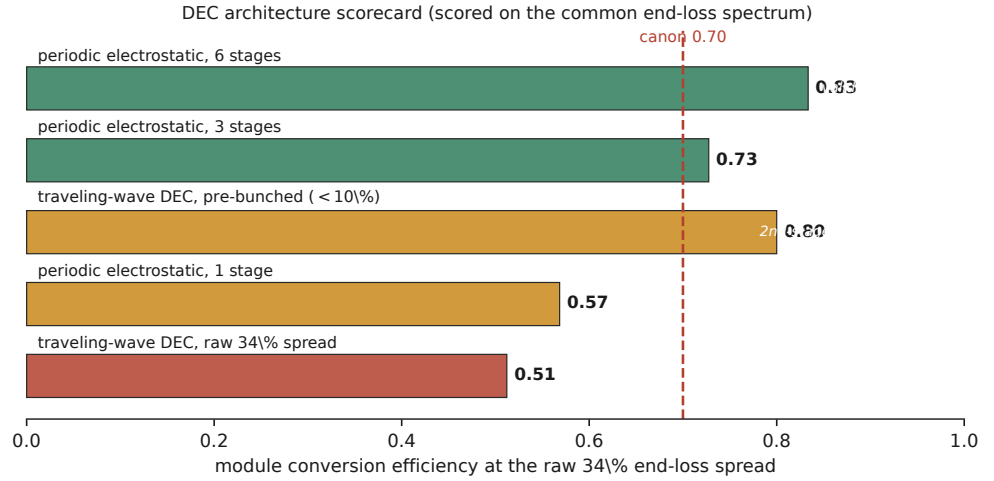


Figure 4. DEC architecture scorecard: module conversion efficiency at the raw 34% end-loss spread. The electrostatic multi-stage collector (green) clears the canonical 0.70 at three and six stages; the TWDEC clears it only pre-bunched (amber) and falls to 0.51 on the raw spread (red).

4.5 Robustness of the baseline

Two sensitivities confirm that the selected 0.70 is conservative. Figure 5 propagates the bound (7): at $M = 6$ the module efficiency tracks the net grid transparency linearly, and even at $\eta_{\text{grid}} = 0.78$ —a pessimistic allowance for optical loss and secondary emission—it stays above the 0.70 floor. Figure 6 contrasts the two stage-placement strategies: energy-optimal (quantile) spacing reaches 0.70 at three stages, whereas uniform-voltage spacing needs ~ 13 stages for the same target—the placement of (8) buys a factor of four in hardware count. This is why an optimally spaced collector, not a naive uniform cascade, is the baseline.

4.6 System efficiency and the frozen gain

Figure 7 decomposes the burner power balance (10) at the frozen operating point (register anchor BU-L1-A10). The axial charged end-loss $P_{\text{ax}} = 2719.7$ MW is routed to the DEC, whose electrical output $\eta_{\text{DEC}} P_{\text{ax}} = 1903.8$ MWe is 54.1% of the gross; the unconverted 30% plus radiation and neutron power are recovered thermally at $\eta_{\text{th}} = 0.45$ for a further 1618.2 MWe. The gross 3522.0 MWe against a recirculating 2672.0 MWe gives $Q_E = 1.318$ and an exhaust-to-electric efficiency of 0.640. The distinction of §1 is explicit in the numbers: $f_{\text{dir}} = 0.4945$ is the *charged* fraction of exhaust, while the DEC’s share of *electricity* is 0.541; the two differ because the thermal path also produces power.

By (11) the output sensitivity is $\partial P_{\text{out}} / \partial \eta_{\text{DEC}} = (1 - 0.45) P_{\text{ax}} \approx 15$ MWe per point of η_{DEC} . The register flags η_{DEC} as a load-bearing input on this basis. The selected electrostatic collector

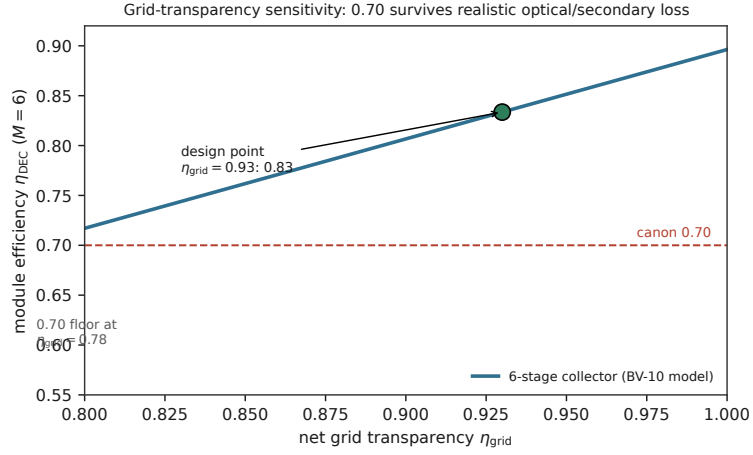


Figure 5. Grid-transparency sensitivity of the six-stage collector (BV-10). The module efficiency scales with the net grid transparency η_{grid} (Eq. (7)); the 0.70 target survives down to $\eta_{grid} \approx 0.78$, so realistic optical and secondary-electron losses do not threaten the frozen assumption.

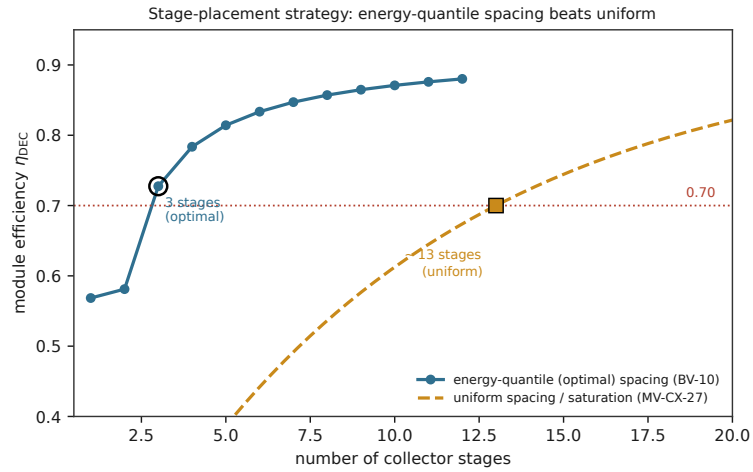


Figure 6. Stage-placement strategy. Energy-quantile (optimal) spacing, the discrete solution of Eq. (8) (BV-10), reaches 0.70 at three stages; uniform-voltage spacing (MV-CX-27) needs ~ 13 stages for the same efficiency—a real bill-of-materials difference set purely by grid placement.

delivers η_{DEC} in the band 0.73–0.83, comfortably above the frozen 0.70, so the burner gain $Q_E = 1.318$ is delivered with headroom rather than at the edge. Two further sensitivities close the loop. Figure 8 shows that the DEC efficiency is essentially flat under plant throttling: as output drops to 16% of full load, φ_i shifts only $\sim 5\%$ ($248.7 \rightarrow 235.1$ keV) and, with fixed collector stages, η_{DEC} falls only from 70.0% to 68.7%—so throttling is not the reason to hold baseload. Figure 9 shows the downstream advantage of an HVDC-native converter: the DEC-direct path reaches 88.4% end-to-end in three stages against 85.4% for a solid-state-transformer-fed path in four (+3.0 points, one fewer stage), and a two-stage DEC-to-bus conversion holds 98.2% [34].

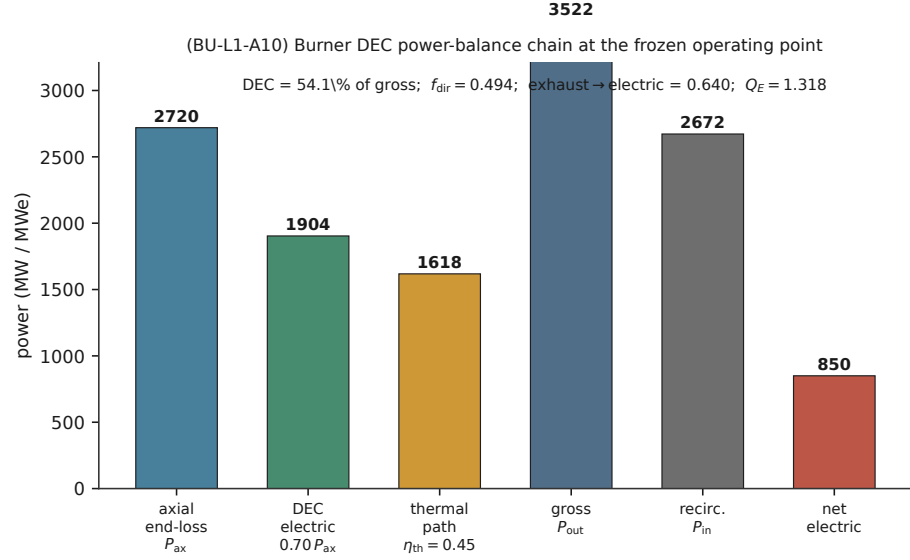


Figure 7. Burner DEC power-balance chain at the frozen operating point (BU-L1-A10), Eq. (10). Axial end-loss $P_{\text{ax}} = 2719.7$ MW; DEC electric 1903.8 MWe (54.1% of gross); thermal recovery 1618.2 MWe; gross 3522.0 MWe; recirculating 2672.0 MWe; net 850 MWe; $Q_E = 1.318$, exhaust-to-electric 0.640.

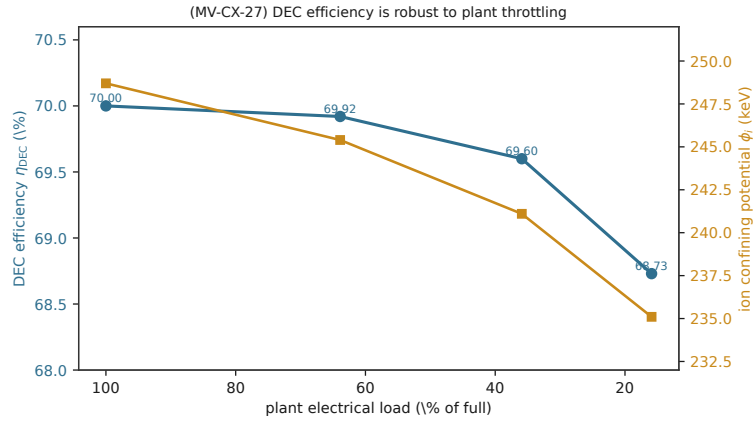


Figure 8. DEC efficiency under plant throttling (MV-CX-27). As electrical load drops from full to 16%, the ion-confining potential φ_i shifts only $\sim 5\%$ and the fixed-stage η_{DEC} falls from 70.0% to 68.7%: the converter is robust to load-following.

4.7 The selected architecture as a schematic

Figure 10 places the selected collector in the machine: the central cell between the 26.49 T plugs, the expander fanning the end-loss stream to reduce its perpendicular energy and space-charge density before the grids, and the periodic electrostatic collector at each end recovering the $f_{\text{dir}} = 0.49$ routed fraction at $\eta_{\text{DEC}} = 0.70$.

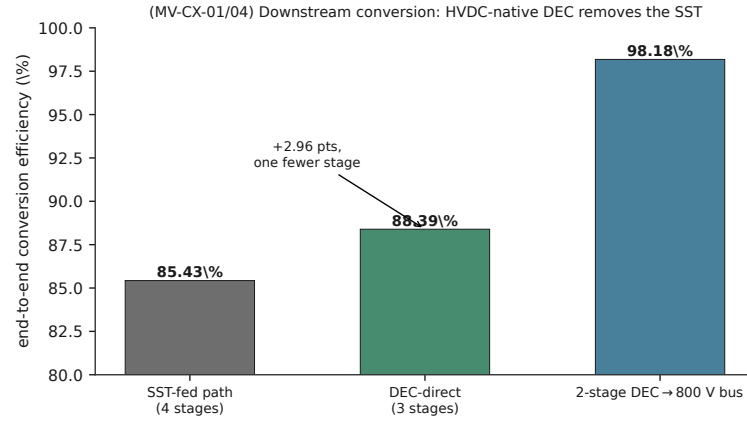


Figure 9. Downstream conversion (MV-CX-01/04). An HVDC-native DEC-direct path reaches 88.4% end-to-end in three stages, beating a solid-state-transformer-fed path (85.4%, four stages) by 3.0 points; the two-stage DEC-to-bus conversion holds 98.2% [34].

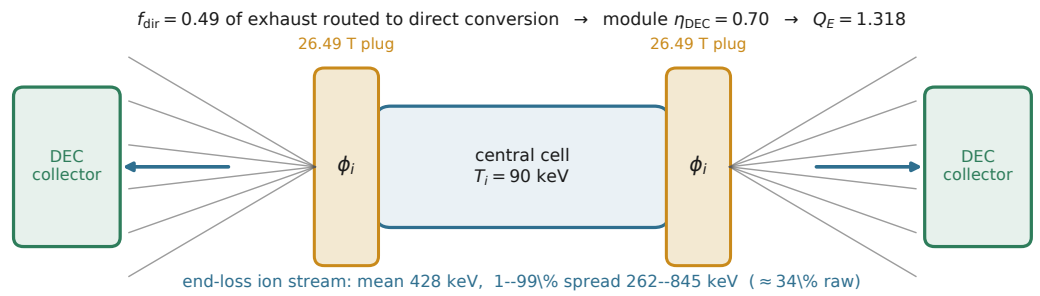


Figure 10. (Schematic.) The tandem-mirror burner and where the DEC sits. The central cell is confined between 26.49 T plugs; the end-loss ion stream (mean 428 keV, 262–845 keV) is expanded and decelerated against the periodic electrostatic collector at each end. The routed fraction $f_{\text{dir}} = 0.49$ is converted at $\eta_{\text{DEC}} = 0.70$, closing $Q_E = 1.318$.

5 Discussion

5.1 Against the experimental and computational record

The selected baseline sits squarely inside the measured DEC record and does not extrapolate beyond it. Single-stage plasma direct converters reached $\eta \approx 0.48$ – 0.50 and multi-stage laboratory collectors 0.65 – 0.80 in the Moir–Barr experiments [3, 4]; the frozen 0.70 sits mid-band, and the six-stage 0.834 is at the upper edge of what has been measured, not above it. The bound (7) explains why: no collector can exceed its own grid transparency, so a claim of $\eta_{\text{DEC}} \rightarrow 1$ would be unphysical, and the honest ceiling here is $\eta_{\text{grid}} = 0.93$. The TWDEC result is likewise consistent with the traveling-wave literature, in which high extraction efficiency is reported specifically for velocity-bunched beams [9, 10]; our contribution is to quantify, in closed form (9), how quickly that efficiency is lost on a raw 34%-spread beam, and thereby to place the TWDEC correctly as a second-stage, tail-converter option rather than a primary. Secondary-electron collectors for the electron channel [11] remain complementary and are not on the critical path to the frozen Q_E .

5.2 Why the simplest architecture wins

The down-select is a case where the simplest architecture wins on the merits. The multi-stage electrostatic collector needs no beam conditioning, meets the target at three stages, and has monotonic headroom; the traveling-wave converter, which is superior for a monoenergetic beam, requires an upstream velocity-buncher to compete on the real exhaust and so imports hardware and complexity that must earn their place. Retaining the TWDEC and the Venetian-blind collector in the portfolio as staged options means the burner keeps an efficiency-upgrade path—a second-stage tail converter behind a buncher, or a higher- η_{grid} ribbon geometry—without any of them being on the critical path to the frozen gain. This is the same design discipline used across the Kronos burner line: freeze the conservative baseline, carry the upgrades as options [28, 29, 30].

5.3 Placement in the burner design

The converter does not stand alone. Upstream, the expander optics that set the beam’s perpendicular energy and space-charge density before the grids are fixed by the end-cell and flux-expansion studies [32, 33]; downstream, the HVDC-native power path that the DEC feeds is fixed by the interface work [34]; and the electrode material that governs grid lifetime is the low-activation, tantalum- and rhenium-free alloy family [31]. The present down-select closes the middle link—the collector architecture itself—and hands a validated $\eta_{\text{DEC}} = 0.70$ module, with a defined upgrade path, to the integrated burner design.

6 Limitations

The efficiencies here are from retarding-field and velocity-acceptance models over the validated end-loss spectrum, not from a full electromagnetic particle-in-cell design of the grids: space-charge limits, beamlet neutralisation, and secondary-electron transport at the 140 MW-class exhaust are engineering constraints on the module’s *power density and thermal* handling that are not captured by the efficiency models. The single open engineering item that follows is grid erosion and lifetime under the sustained 428 keV end-loss flux (AG-CX-10): it sets the collector maintenance interval and the electrode-material choice, and it is the next scheduled run—a GPU-fleet velocity-space PIC of the grids with WarpX or Smilei [20, 21] coupled to the electrode-alloy sputtering response [31]. It refines the module’s durability, not its efficiency, and does not change the architecture selection.

7 Conclusion

Against a common 428 keV end-loss spectrum with a raw 34% energy spread, the direct-energy converter for the Kronos D - ^3He tandem-mirror burner is a periodic multi-stage electrostatic collector. Formalised as a retarding-field quantile-optimisation problem, it meets the canonical module efficiency $\eta_{\text{DEC}} = 0.70$ at three stages (0.727), reaches 0.834 at six, and is bounded above only by its grid transparency $\eta_{\text{grid}} = 0.93$. The traveling-wave converter, whose extraction efficiency falls as $\exp[-\sigma_E^2/(8\sigma_{\text{acc}}^2)]$, delivers 0.75 – 0.80 on a narrow pre-bunched beam but only 0.51 on the raw spread, and is retained as a second-stage tail option behind a velocity-buncher; the Venetian-blind and periodic-electrostatic-plug collectors are the staged build. Applied to the routed fraction $f_{\text{dir}} = 0.49$, the selected architecture delivers the frozen engineering gain $Q_E = 1.318$ with headroom, is robust to plant throttling and grid-transparency loss, and feeds an HVDC-native downstream path. Grid erosion and lifetime under the end-loss flux is the one qualifying item that remains.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim. Converter and spectrum computations ran on the Kronos NVIDIA GPU HPC campaign alongside an in-house HPC node for the reduced-order and analytic layers.

Reproducibility. The DEC-architecture results are frozen anchors in the Kronos Physics De-Risking Register [27] (doi:10.5281/zenodo.22133057): the periodic-electrostatic multi-stage validation and end-loss spectrum (BV-10), the traveling-wave acceptance model (AG-CX-12), the throttle-robustness and stage-strategy models (MV-CX-27), the downstream conversion chain (MV-CX-01, MV-CX-04), and the burner DEC power-balance chain (BU-L1-A10). Each is regenerable from the archived converter models and the figure-generation script deposited with this paper. This paper is archived at doi:10.5281/zenodo.22645935 and its official version is maintained at <https://www.kronosfusionenergy.com/publications/direct-energy-converter-architectures-for-a-d-he-tandem.html>.

Data availability

The collector efficiency curves, the end-loss spectrum, and the power-balance decomposition used for Figures 1–10 are archived with the deposit at doi:10.5281/zenodo.22645935, which references the Kronos Physics De-Risking Register [27] (doi:10.5281/zenodo.22133057), the burner design paper [28], and the direct-energy-conversion baseline [29]. All quantitative values trace to the register by the serial identifiers cited inline. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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